

Statistic vs. Parameter: What's the Difference?

There are two important terms in the field of inferential statistics that you should know the difference between: **statistic** and **parameter**.

A **statistic** is a number that describes some characteristic of a sample.

A **parameter** is a number that describes some characteristic of a population.

population represents every possible individual element that you're interested in measuring, while a sample simply represents a portion of the population.

For example, you may be interested in identifying the mean height of palm trees in Florida. There might be tens of thousands of palm trees around the state, which means it would be virtually impossible to go around and measure the height of every single one.

Instead, you may select a random sample of 100 palm trees and find the mean height of the trees in just that sample. Suppose the mean turns out to be 36 feet.

In this example, the population is every palm tree in Florida. The sample is the group of 100 trees that we randomly selected.

The **statistic** is the mean height of the trees in our sample – 36 feet.

The **parameter** is the true mean height of *all* palm trees in Florida, which is unknown since we will never be able to measure every single palm tree in the state.

The parameter is the value that we're actually interested in measuring, but the statistic is the value that we use to estimate the value of the parameter since the statistic is so much easier to obtain.

Commonly Used Statistics and Parameters

Note that we write parameters and statistics using different symbols.

Measurement	Sample statistic	Population parameter
Mean	$\bar{x}$	μ (mu)
Standard deviation	s	σ (sigma)
Variance	s^2	σ^2 (sigma squared)
Proportion	p	π (pi)
Correlation	r	ρ (rho)
Regression coefficient	b	β (beta)

In any problem, we are always interested in measuring the population parameter. However, it's often too time-consuming, too costly, or simply not possible to actually measure every single individual element in the population, which is why we instead calculate a sample statistic and use that statistic to estimate the true population parameter.

Nerd notes:

To ensure that our sample statistic is a good estimate for the true population parameter, we need to make sure that we obtain a representative sample – a sample in which the characteristics of the individuals closely match the characteristics of the overall population.

Statistic vs. Parameter: Practice Problems

The following practice problems will help you gain a better understanding of the difference between statistics and parameters.

First, read the problem. Then, try to identify the statistic and the parameter in each problem. The correct answer will be listed below each problem so that you can check your work.

Problem #1

A researcher would like to find the mean wingspan of a certain bird species. She collects a random sample of 50 birds, measures the wingspan of each bird, and finds that the mean wingspan is 15.6 inches.

Answer: The **parameter** that the researcher is interested in measuring is the mean wingspan for the entire population of this particular bird species. The **statistic** is the sample mean, which turns out to be 15.6 inches.

Problem #2

An election council wants to understand what proportion of adults in a certain city are in favor of a particular tax law. They obtain a random sample of 1,000 adults and find that 34% are in favor of the law.

Answer: The **parameter** that the council is interested in measuring is the proportion of all adults in the city who are in favor of the tax law. The **statistic** is the sample proportion, which turns out to be 34%.

Problem #3

A team of economists wants to estimate the standard deviation of incomes among adults in a certain country. They obtain a random sample of 10,000 adults and find that the standard deviation among their incomes is \$12,500.

Answer: The **parameter** that the team of economists is interested in measuring is the standard deviation of incomes among all adults in the country. The **statistic** is the sample standard deviation, which turns out to be \$12,500.

Problem #4

A researcher wants to estimate the mean coffee consumption of students at a particular university. He obtains a random sample of 200 students and finds that the mean coffee consumption is 2.2 cups per day per student.

Answer: The **parameter** that the researcher wants to measure is the mean coffee consumption of all students at this university. The **statistic** is the sample mean, which turns out to be 2.2 cups per day per student.